

Questions with Answer Keys

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Q1: 16 March (Shift 1) - Numerical

If $\lim_{x \rightarrow 0} \frac{ae^x - b \cos x + ce^{-x}}{x \sin x} = 2$, then $a + b + c$ is equal to _____

Q2: 17 March (Shift 1) - Single Correct

The value of $\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(x - [x]^2) \cdot \sin^{-1}(x - [x]^2)}{x - x^3}$, where $[x]$ denotes the greatest integer $\leq x$ is :

- (1) π
- (2) 0
- (3) $\frac{\pi}{4}$
- (4) $\frac{\pi}{2}$

Q3: 17 March (Shift 2) - Single Correct

The value of $\lim_{n \rightarrow \infty} \frac{[r] + [2r] + \dots + [nr]}{n^2}$, where r is non-zero real number and $[r]$ denotes the greatest integer less than or equal to r , is equal to :

- (1) $\frac{r}{2}$
- (2) r
- (3) $2r$
- (4) 0

Q4: 17 March (Shift 2) - Single Correct

The value of the limit $\lim_{\theta \rightarrow 0} \frac{\tan(\pi \cos^2 \theta)}{\sin(2\pi \sin^2 \theta)}$ is equal to :

- (1) $-\frac{1}{2}$
- (2) $-\frac{1}{4}$
- (3) 0
- (4) $\frac{1}{4}$

Q5: 18 March (Shift 1) - Single Correct

If $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - \tan^{-1} x}{3x^3}$ is equal to L , then the value of $(6L + 1)$ is

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(2) $\frac{1}{2}$

Answer Key**Q1 (4)****Q2 (4)****Q3 (1)****Q4 (1)****Q5 (4)**